STUDENT REFERENCE & INTERNAL USE NOTICE

This protocol documents standard antimicrobial susceptibility testing procedures used in HOPE Lab research and is provided to lab members for training and internal reference. External users should independently assess method suitability, verify current standards, and obtain required biosafety approvals. This document does not replace clinical diagnostic guidelines.

01. TITLE

Antimicrobial Susceptibility Testing (AST) Protocol: Kirby-Bauer Disk Diffusion Method

02. PRINCIPLE

Antimicrobial susceptibility testing (AST) measures the ability of a microorganism to grow in the presence of specific antimicrobial agents under standardized conditions. In the Kirby-Bauer disk diffusion method, paper disks impregnated with precise concentrations of antibiotics are placed on Mueller-Hinton agar inoculated with a standardized bacterial suspension (Hossain, 2024). The antibiotic diffuses radially outward, forming a concentration gradient. Where the local concentration exceeds the Minimum Inhibitory Concentration (MIC), bacterial growth is inhibited, producing a clear zone of inhibition. Zone diameters are measured in millimeters and compared against CLSI/EUCAST reference breakpoints to categorize the organism as Susceptible (S), Intermediate (I), or Resistant (R).

Learning Outcomes

- Standardize bacterial suspensions to a 0.5 McFarland standard visually and spectrophotometrically.
- Inoculate Mueller-Hinton agar to achieve uniform, confluent bacterial lawns.
- Apply antimicrobial disks accurately with correct geometric spacing.
- Measure zones of inhibition precisely using digital calipers or precision scales.
- Interpret AST phenotypic profiles using CLSI/EUCAST clinical breakpoint criteria.

03. OBJECTIVE & SCOPE

Objective: To provide a reproducible, standardized procedure for evaluating antibiotic resistance phenotypes in bacterial isolates from food, environmental, clinical, and probiotic sources (Ali et al., 2021; Paul et al., 2024).

Scope: Applicable to rapidly growing aerobic and facultative anaerobic bacterial pathogens and gut commensals (e.g., *Salmonella* spp., *E. coli*, *Staphylococcus* spp., and lactic acid bacteria). Excludes obligate anaerobes and slow-growing fastidious species requiring specialized media or atmosphere.

04. CORE MECHANISMS OF ANTIMICROBIAL ACTION

- **Cell Wall Synthesis Inhibition:** Target peptidoglycan cross-linking (e.g., Penicillins, Cephalosporins), inducing osmotic lysis.
- **Membrane Integrity Disruption:** Alter membrane permeability (e.g., Polymyxins, Daptomycin), causing intracellular leakage.
- **Protein Synthesis Inhibition:** Bind 30S or 50S ribosomal subunits (e.g., Aminoglycosides, Tetracyclines), arresting translation.
- **Nucleic Acid Synthesis Inhibition:** Inhibit DNA gyrase/topoisomerase IV (e.g., Fluoroquinolones) or RNA polymerase (e.g., Rifampicin).
- **Metabolic Pathway Inhibition:** Competitive inhibition of folic acid biosynthesis (e.g., Sulfonamides, Trimethoprim).

05. MATERIALS & EQUIPMENT

Equipment & Software

- Calibrated Incubator (35 ± 2°C)
- Spectrophotometer / Densitometer (OD_{600})
- Vortex Mixer & Inoculation Loops
- Digital Precision Caliper / Ruler

Reagents & Consumables

- Mueller-Hinton Agar (MHA) plates (4 mm depth)
- Standardized Antibiotic Disks (CLSI/EUCAST compliant)
- Sterile 0.85% Saline Solution
- 0.5 McFarland Turbidity Standard
- Sterile Cotton Swabs & Forceps
- Non-selective media (Nutrient Agar / TSA)

06. STEP-BY-STEP PROCEDURE

Step 1 : Preparation of Bacterial Suspension (Inoculum)

Pick 3–5 well-isolated colonies from an 1 8–24 h non-selective culture plate using a sterile loop. Emulsify in 0.85% sterile saline. Adjust turbidity to equal 0.5 McFarland standard (approx. 1 –2 × 1 0^{8} CFU/mL for *E. coli*) using a spectrophotometer (OD_{600} = 0.08–0.1 0).

Step 2: Inoculation of Mueller-Hinton Agar Plate

Within 1 5 minutes of suspension adjustment, dip a sterile cotton swab into the inoculum. Express excess liquid by pressing the swab against the inside tube wall. Streak the swab evenly across the entire MHA surface in three directions, rotating the plate 60° each time. Pass the swab around the agar rim to ensure full lawn coverage. Allow 3–5 minutes for surface moisture absorption.

Step 3: Application of Antibiotic Disks

Using sterile forceps or an automated disk dispenser, place antimicrobial disks onto the inoculated agar. Press each disk lightly to ensure complete contact. Space disks at least 24 mm apart (center-to-center) and at least 1 5 mm from the plate edge (maximum 6–8 disks per 1 00 mm plate).

Step 4: Incubation

Invert plates within 1 5 minutes of disk application and place in an incubator maintained at 35 ± 2°C for 1 6–1 8 hours in an aerobic atmosphere.

Step 5: Measurement of Inhibition Zones

Examine plates against a non-reflecting black background with direct illumination. Measure the complete diameter of clear growth inhibition zones (including disk diameter) in millimeters using digital calipers.

Step 6: Interpretation & Result Recording

Compare measured zone diameters against official standard organism-specific breakpoint tables:

- **Susceptible (S):** Infection responds to standard dosing regimen.
- **Intermediate (I):** Efficacy uncertain; buffer zone to prevent technical misclassification.
- **Resistant (R):** High probability of therapeutic failure.

07. CRITICAL FACTORS AFFECTING AST RESULTS

Variable	Effect of Improper Control	Required Control Standard
Inoculum Density	Heavy inoculum → smaller zones; light inoculum → artificially large zones	Standardize to 0.5 McFarland immediately before inoculation
Agar Depth	> 4 mm → false resistance (small zones); < 4 mm → false susceptibility	Maintain uniform 4.0 ± 0.5 mm depth (~25 mL per 90 mm dish)
Medium pH & Divalent Cations	pH < 7.2 reduces macrolide activity; Ca^{2+}/Mg^{2+} alter aminoglycoside/tetracycline diffusion	Verify MHA pH (7.2–7.4 at 25°C) and batch-controlled cation levels

Disk Storage & Potency	Humidity & heat degrade labile drugs (e.g., β-lactams), causing false resistance	Store disk cartridges at -20°C; warm to room temp before opening
Disk Spacing	Overlapping inhibition zones prevent accurate diameter measurement	Maintain minimum 24 mm center-to-center distance between disks

08. QUALITY CONTROL & TROUBLESHOOTING

Quality Control (QC): Test reference strain controls (e.g., *E. coli* ATCC 25922, *S. aureus* ATCC 25923) alongside test runs. Verify that QC zone diameters fall strictly within standard CLSI acceptance ranges.

Problem	Likely Cause	Corrective Action
No or faint growth	Non-viable culture, overheating loop, or low incubator temperature	Re-subculture test strain; verify incubator temperature (35 ± 2°C)
Inconsistent QC zone sizes	Inoculum density variance, expired disks, or variable agar depth	Recalibrate densitometer/McFarland standard; replace disk batch
Colonies within zone	Mixed culture or emergence of resistant mutant subpopulations	Check culture purity by subculturing single colony and repeating test

09. BIOSAFETY & ADDITIONAL CONSIDERATIONS

- **Biosafety:** Perform all manipulations involving potential pathogens inside a Biosafety Cabinet using BSL-2 precautions. Dispose of contaminated agar plates in biohazard bags for autoclaving.
- **Clinical & Ecological Context:** Phenotypic AST screening must be combined with organism identification and molecular AMR gene detection for comprehensive One Health reporting.

10. REFERENCES & RELATED DOCUMENTS

1. Ali F, Silvy TN, Hossain TJ, Uddin MK, Uddin MS. (2021). Prevalence and antimicrobial resistance phenotypes of *Salmonella* species recovered at various stages of broiler operations in Hathazari, Bangladesh. *International Journal of One Health*, 7(2), 158–164. doi:10.4202/IJOH.2021.158-164
2. Hossain TJ. (2024). Methods for screening and evaluation of antimicrobial activity: a review of protocols, advantages and limitations. *European Journal of Microbiology and Immunology*, 14(2), 97–115. doi:10.1556/1886.2024.00035
3. Paul S, Hossain TJ, Ali F, Hossain ME, Chowdhury T, Faisal IK, Ferdouse J. (2024). Assessment of the in-vitro probiotic efficacy and safety of *Pediococcus pentosaceus* L1 and *Streptococcus thermophilus* L3 isolated from Laban, a popular fermented milk product. *Archives of Microbiology*, 206, 82. doi:10.1007/s00203-023-03812-5